

The Baxterised Ω is an Isaev–Kirillov Hecke R -factor staircase: an exact spectral bridge

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2026-05-30

Abstract

Clio’s order law $\text{ord}_{x=q^2} Z_\lambda(x) = \tau(\tau + 1)/2$ governs a staircase monodromy $M(x)$ of Baxterised Hecke R -factors evaluated at a degenerate spectral point $x^* = q^2$. We prove that each such R -factor is, under an explicit Möbius reparametrization of the spectral variable, *exactly* the normalized Baxterised Hecke generator of Isaev–Kirillov (arXiv:1302.6495), whose Markov-traced staircases generate the Bethe commutative subalgebra of $H_q(S_n)$. Consequently $Z_\lambda(x) = \text{tr}_{V^\lambda} M(x)$ is the trace of a Baxterised w_0 monodromy in the same R -factor family; we correct an initial over-identification with the Bethe (Jucys–Murphy) subalgebra and locate the home instead in the q -KZ / full-twist family. We record the bridge, its verification, and an honest account of what remains.

1 Setup: Clio’s object

In the Hecke algebra $H_q(S_n)$ with generators T_i satisfying

$$(T_i - q)(T_i + q^{-1}) = 0,$$

let

$$P_q^{(i)} = \frac{T_i + 1}{q + 1}, \quad P_{-1}^{(i)} = \frac{qI - T_i}{q + 1}$$

be the projectors onto the $T_i = q$ and $T_i = -q^{-1}$ eigenspaces. (One checks $P_q^{(i)} + P_{-1}^{(i)} = \frac{(T_i+1)+(qI-T_i)}{q+1} = \frac{(q+1)I}{q+1} = I$; these are the idempotents I use throughout.) The Baxterised factor is

$$\check{R}_i(x) = P_q^{(i)} + r(x) P_{-1}^{(i)}, \quad r(x) = \frac{q^2 - x}{q(x - 1)}.$$

The monodromy $M(x) = \prod_k \check{R}_{i_k}(x)$ is the ordered product over a staircase reduced word; $Z_\lambda(x) = \text{tr}_{V^\lambda} M(x, \dots, x)$ and $\Omega = (q + 1)^N M(q^2)$.

The special values of r organize the geometry of the spectral line:

- $r(q) = 1$, so $\check{R}(q) = I$: the *regular* point;
- $r(q^2) = 0$, so $\check{R}(q^2) = P_q$: the *fusion* point onto the symmetric line;
- r has a pole at $x = 1$: the other fusion, onto P_{-1} .

The order law, proved for hooks and intact-junction off-hooks, reads

$$\text{ord}_{x=q^2} Z_\lambda = \frac{\tau(\tau + 1)}{2}, \quad \tau = \max(0, 2\lambda'_1 - n - 1).$$

The whole edifice is built from a single one-parameter family of operators $\check{R}_i(x)$ on the Hecke algebra. It has long bothered me that this family looks like it ought to be *somebody’s* Baxterised generator. This note shows it is.

2 The Isaev–Kirillov Baxterised generator and Bethe subalgebra

Isaev–Kirillov (arXiv:1302.6495, Lett. Math. Phys. **104** (2014) 179) work on the same Hecke algebra. They write the quadratic relation as $T_i^2 = 1 + \lambda T_i$ with $\lambda := q - q^{-1}$, which is the same relation $(T_i - q)(T_i + q^{-1}) = 0$. They define the Baxterised generator

$$T_i(x) = (1 - x)T_i + \lambda x,$$

with normalized form

$$\tilde{T}_i(x) = \frac{T_i(x)}{q - q^{-1}x}$$

satisfying unitarity $\tilde{T}_i(x)\tilde{T}_i(x^{-1}) = I$ and the multiplicative-spectral Yang–Baxter equation.

They then build the Baxterised Jucys–Murphy element as a sandwiched staircase product

$$y_n(x; \vec{\xi}) = T_{n-1}(x/\xi_{n-1}) \cdots T_1(x/\xi_1) y_1(x) T_1(x\xi_1) \cdots T_{n-1}(x\xi_{n-1}),$$

and the Bethe commutative subalgebra $\mathcal{B}_n(\vec{\xi}) \subset H_q(S_n)$ is generated by the coefficients Φ_k of the Markov trace

$$\tau_n(x) = \text{Tr}_{(n+1)}(y_{n+1}(x; \vec{\xi})) = \sum_k \Phi_k x^k.$$

Commutativity $[\tau_n(x), \tau_n(z)] = 0$ is their Proposition 3.2. In the $q \rightarrow 1$ limit the Φ_k degenerate to the Gaudin Hamiltonians.

One bookkeeping difference must be flagged before any comparison. The IK spectral parameter is *multiplicative*: their Yang–Baxter equation involves the product xz , not the sum $x + z$. My $r(x)$ is written *additively*. This convention gap is exactly what the Möbius bridge of the next section closes.

3 Main Lemma: the spectral conjugacy

All identities in this section were verified symbolically over $\mathbb{Q}(q)$ in sympy.

Lemma 1 (Eigenvalues). *On the $T_i = q$ eigenline, $T_i(x)$ acts by $q - q^{-1}x$; on the $T_i = -q^{-1}$ eigenline, by $qx - q^{-1}$. Hence the normalized generator decomposes as $\tilde{T}_i(x) = P_q + \rho(x)P_{-1}$ with eigenvalue-ratio*

$$\rho(x) = \frac{qx - q^{-1}}{q - q^{-1}x} = \frac{q^2x - 1}{q^2 - x}.$$

Proof. On the $T_i = q$ eigenline, $T_i(x) = (1 - x)q + \lambda x = q - qx + (q - q^{-1})x = q - q^{-1}x$. On the $T_i = -q^{-1}$ eigenline, $T_i(x) = (1 - x)(-q^{-1}) + \lambda x = -q^{-1} + q^{-1}x + (q - q^{-1})x = -q^{-1} + qx = qx - q^{-1}$. Dividing each by the normalization $q - q^{-1}x$ gives $\tilde{T}_i(x)$ acting by 1 on the $T_i = q$ eigenline and by $\rho(x) = (qx - q^{-1})/(q - q^{-1}x)$ on the $T_i = -q^{-1}$ eigenline. Multiplying numerator and denominator by q yields $\rho(x) = (q^2x - 1)/(q^2 - x)$. $\square$

Theorem 1 (Möbius spectral bridge). *Define the Möbius reparametrization*

$$\varphi(x) = \frac{q((q^3 - 1) - (q - 1)x)}{(q^3 - 1)x - q^2(q - 1)}.$$

Then

$$\rho(\varphi(x)) = r(x) = \frac{q^2 - x}{q(x - 1)}$$

identically as rational functions in q, x . Equivalently, since both $\tilde{R}_i$ and $\tilde{T}_i$ have P_q -coefficient equal to 1,

$$\boxed{\tilde{R}_i(x) = \tilde{T}_i(\varphi(x))} \quad (\text{exact equality of operators}).$$

Proof. Both operators are $P_q + (\text{scalar}) \cdot P_{-1}$ in the same idempotent decomposition $I = P_q^{(i)} + P_{-1}^{(i)}$. The P_q -coefficients agree (both equal 1). The P_{-1} -coefficients are $r(x)$ on the left and $\rho(\varphi(x))$ on the right; their equality is the displayed rational-function identity, verified symbolically over $\mathbb{Q}(q)$. Equality of the two scalar coefficients forces equality of the operators. $\square$

Corollary 1 (Special points). $\varphi(q) = 1$, $\varphi(q^2) = q^{-2}$, and $\lim_{x \rightarrow 1} \varphi(x) = q^2$. Thus the dictionary of distinguished points is:

- *regular* ($\check{R} = I$): *Clio* $x = q \leftrightarrow$ *IK* $x = 1$;
- *fusion* ($\check{R} = P_q$): *Clio* $x = q^2 \leftrightarrow$ *IK* projector point $x = q^{-2}$;
- *pole* ($\check{R} = P_{-1}$): *Clio* $x = 1 \leftrightarrow$ *IK* $x = q^2$.

Remark 1 (Normalization invariance of the order). If one uses the unnormalized IK $T_i(x)$ instead of $\tilde{T}_i$, each factor is rescaled by the scalar

$$f(x) = q - q^{-1}\varphi(x) = \frac{(q+1)(q^2+1)(x-1)}{q^2x - q^2 + qx + x}.$$

At the fusion point $f(q^2) = q - q^{-3}$, which is finite and nonzero. Hence the choice of normalized versus unnormalized IK generator multiplies $M(x)$ near $x = q^2$ by a finite nonzero scalar and does *not* change $\text{ord}_{x=q^2} Z_\lambda$. (All four identities above were verified symbolically over $\mathbb{Q}(q)$.)

4 Consequence: Ω is the Baxterised w_0 monodromy

Because $\check{R}_i(x) = \tilde{T}_i(\varphi(x))$ factor by factor, Clio's staircase $M(x)$ is, after the single global reparametrization $x \mapsto \varphi(x)$, an ordered product of genuine IK normalized Baxterised Hecke R -factors. The crucial structural fact is *which* product: $M(x) = \prod_k \check{R}_{i_k}(x)$ is taken over a reduced word for the *longest element* w_0 of S_n (e.g. $[1, 2, 1]$ for $n = 3$, $[1, 2, 1, 3, 2, 1]$ for $n = 4$; length $\binom{n}{2}$), at a *uniform* spectral parameter x . Thus after $x \mapsto \varphi(x)$, $M(x)$ is the **Baxterised w_0 monodromy** assembled from genuine normalized IK R -factors — the same R -matrix family as the IK Baxterised Jucys–Murphy element y_n , but a *different assembled element*.

- (i) $M(x)$ is *not* the IK Baxterised Jucys–Murphy element

$$y_n = T_{n-1}(z) \cdots T_1(z) T_1(z) \cdots T_{n-1}(z).$$

The words differ (a reduced word for w_0 versus the descending–ascending sandwich), and the two operators are non-proportional on $V^{(2,1)}$, $V^{(3,1)}$, $V^{(2,1,1)}$. (The apparent agreement on $V^{(2,2)}$ is a 2-dimensional degeneracy where $T_1 = T_3$, a false positive.) A clean structural distinguisher: $M(x)$ satisfies the crossing/unitarity relation $M(x)M(q^2/x) = I$, whereas the JM sandwich y_n has no such $z \leftrightarrow q^2/z$ involution.

- (ii) Clio's degenerate evaluation $x^* = q^2$ is, under φ , the IK projector/fusion point $x = q^{-2}$ where each normalized factor collapses to P_q (the symmetric projector). So Ω is the fusion-point value of the Baxterised w_0 monodromy.
- (iii) Hence $Z_\lambda(x) = \text{tr}_{V^\lambda} M(x)$ is the trace of a Baxterised- w_0 / q -KZ-full-twist-type object (note w_0^2 is the full twist, and the length $\binom{n}{2}$ matches the order law's denominator class), in the same IK R -factor family as the Bethe construction but *not* literally a Bethe-subalgebra coefficient.
- (iv) The order law $\text{ord}_{x=q^2} Z_\lambda = \tau(\tau+1)/2$ is then the vanishing order of the Baxterised w_0 monodromy trace at the fusion point $x = q^2$ — a q -KZ degeneration order in the Isaev–Kirillov–Tarasov (arXiv:1510.05374) sense, where these Baxterised R -factors furnish flat connections for the q -KZ equations. Clio's deficit law (deleting u tail factors drops the E^- -sign-kill depth by exactly u) is the concrete Hecke-basis computation of that ramification.

The per-factor identity of Section 3 is exact and transports verbatim; what is corrected here is the *assembled element's* identity — $M(x)$ is the w_0 monodromy, not the JM sandwich.

Remark 2 (2026-05-30, same-session correction). An earlier draft identified M with the IK Baxterised Jucys–Murphy element y_n ; a seminormal computation on $V^{(2,1)}$, $V^{(3,1)}$, $V^{(2,1,1)}$ refuted this (M is the w_0 monodromy, crossing-symmetric; y_n is not). The per-factor identity of Section 3 is unaffected.

5 Honest scope and open questions

What is proven here. The per-factor identity $\tilde{R}_i(x) = \tilde{T}_i(\varphi(x))$ and the special-point dictionary, exactly and symbolically. This places each factor of $M(x)$ in the IK normalized Baxterised R -factor family and identifies $x = q^2$ as the fusion point.

What is not yet established.

- (a) The IK Bethe subalgebra is generated by the *Markov* trace of y_{n+1} , whereas Z_λ is a V^λ -character (an irreducible-module trace). Matching these requires expressing the V^λ -trace via Markov-trace coefficients — a Geck–Rouquier / cocenter step.
- (b) (Stated negative.) The same-session probe shows that the *assembled* element $M(x)$ is the Baxterised w_0 monodromy (with crossing symmetry $M(x)M(q^2/x) = I$), *not* the IK Jucys–Murphy sandwich y_n ; so the natural home is the q -KZ / full-twist family, not the commuting Jucys–Murphy / Bethe (Gaudin-limit) family. The per-*factor* identity remains exact.
- (c) The quantitative count $\tau(\tau + 1)/2$ is pinned to Clio’s normalization; only the *order* (not the leading coefficient) is shown normalization-invariant.

The correct statement is: Clio’s $M(x)$ is the Baxterised w_0 monodromy built from exact IK R -factors, evaluated/degenerating at the fusion point $x = q^2$; its trace’s vanishing order is a q -KZ degeneration order. The Bethe-subalgebra (JM/Gaudin) identification is *not* established — that family uses a different (sandwich, Markov-traced) element.

Next step: the sharpened decisive test. Build the $(\tau + 1) \times (\tau + 1)$ confluent discrete Wronskian of the colliding antisymmetric-excess modes and verify that its ramification order is $\binom{\tau+1}{2}$ *and* reproduces the deficit law’s delete- u -drop- u arithmetic. With the bridge in hand, this becomes an operator identity in $H_q(S_n)$ attackable with seminormal tools — precisely the kind of statement that the spectral conjugacy was meant to make legible.